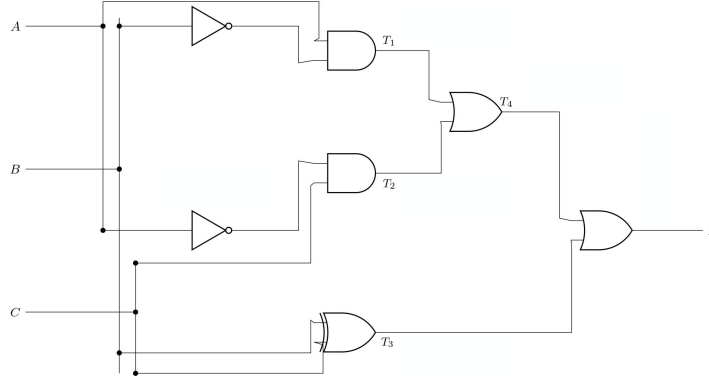


Practice Final Exam — Answer Key
CSCI 160: Computer Architecture 1

Part I: Multiple Choice. 1. B 2. A 3. B 4. B 5. C

Part II: Show Work

1. Circuit with labeled gates: $T_1 = AB'$, $T_2 = A'C$, $T_3 = B \oplus C$, $T_4 = T_1 + T_2$, and $F = T_4 + T_3$.



By substitution

$$F(A, B, C) = AB' + A'C + (B \oplus C) = \sum m(1, 2, 3, 4, 5, 6).$$

Truth table (row order $ABC = 000 \dots 111$):

ABC	000	001	010	011	100	101	110	111
$T_1 = AB'$	0	0	0	0	1	1	0	0
$T_2 = A'C$	0	1	0	1	0	0	0	0
$T_3 = B \oplus C$	0	1	1	0	0	1	1	0
$T_4 = T_1 + T_2$	0	1	0	1	1	1	0	0
F	0	1	1	1	1	1	1	0

2. Subtraction $A - B$, $A = 1010$ (10), $B = 0110$ (6). Mode $M = 1$, so $C_0 = 1$ and each XOR inverts B : $B \rightarrow 1001$. Column by column ($i = 0 \rightarrow 3$):

Position i	3	2	1	0
Carry in C_i	0	1	1	1
A_i	1	0	1	0
B_i (after XOR)	1	0	0	1
Sum S_i	0	1	0	0
Carry out C_{i+1}	1	0	1	1

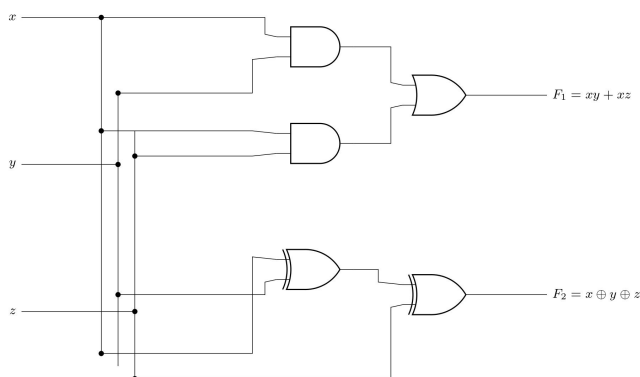
Result $S_3S_2S_1S_0 = 0100 = 4$. End carry $C_4 = 1$: no borrow, so $A \geq B$ (indeed $10 - 6 = 4$).

3. Outputs (row order $xyz = 000 \dots 111$):

xyz	000	001	010	011	100	101	110	111
F_1	0	0	0	0	0	1	1	1
F_2	0	1	1	0	1	0	0	1

$$F_1 = \sum m(5, 6, 7) = xu + xz, \quad F_2 = \sum m(1, 2, 4, 7) = x \oplus y \oplus z.$$

Logic diagram:

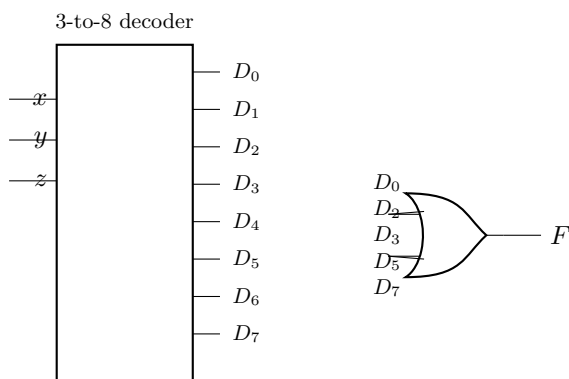


4. BCD $9 + 8$. Raw binary sum $9 + 8 = 17 = 10001$, so $K = 1$, $Z_8Z_4Z_2Z_1 = 0001$. Correction $C = K + Z_8Z_4 + Z_8Z_2 = 1 + 0 + 0 = 1$ (the K term fires), so add 0110:

$$10001 + 0110 = 10111 \Rightarrow \text{carry } 1, \text{ digit } 0111 = 7.$$

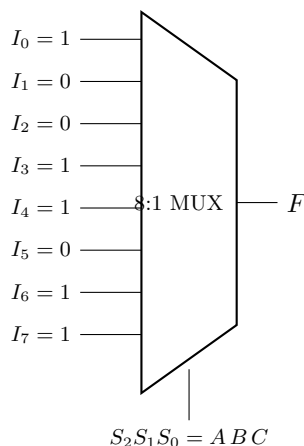
Answer: BCD 00010111 = 17.

5. Decoder puts each minterm on its own line ($D_i = m_i$); OR the lines in the minterm list. For $F(x, y, z) = \sum m(0, 2, 3, 5, 7)$: OR together D_0, D_2, D_3, D_5, D_7 .



6. $F(A, B, C) = \sum m(0, 3, 4, 6, 7)$ on an 8-to-1 MUX, $S_2S_1S_0 = A, B, C$. Data inputs = truth-table column:

$$I_0 = 1, I_1 = 0, I_2 = 0, I_3 = 1, I_4 = 1, I_5 = 0, I_6 = 1, I_7 = 1.$$



7. Subtraction $A - B$, $A = 0111 (7)$, $B = 0010 (2)$, $M = 1$. So $C_0 = 1$ and $B \rightarrow 1101$ (each bit inverted).

- | | |
|--|---|
| (a) $A_0 A_1 A_2 A_3 = 1, 1, 1, 0$ | (b) $B_0 B_1 B_2 B_3 = 1, 0, 1, 1$ (after XOR) |
| (c) $C_0 C_1 C_2 C_3 C_4 = 1, 1, 1, 1, 1$ | (d) $S_0 S_1 S_2 S_3 = 1, 0, 1, 0$ ($S = 0101 = 5$) |
| (e) $V = C_3^{\text{in}} \oplus C_4^{\text{out}} = 1 \oplus 1 = 0$ | (f) No overflow. |

Result $7 - 2 = 5$; $C_4 = 1$ confirms $A \geq B$.